

IPL_Analytics_DW

1) Architecting a High-Performance, Multi-Tier Sports Intelligence Warehouse using Recursive SQL Foundations

System Architecture Overview

Phase 1: Raw Bulk Data Ingestion Layer (Bronze Staging)

Action: Moves raw files (`Matches.csv`, `Deliveries.csv`) directly into SQL Server staging tables.

Format: Every column is set as text (`VARCHAR`) so the load never crashes due to format changes or dirty data.

Phase 2: Algorithmic Row Realignment & Quality Cleansing (Silver Transformations)

Action: Uses a Recursive CTE to hunt down rows twisted out of line by rogue commas inside text fields.

Format: Shifts displaced columns back into place, trims extra whitespace, and casts text numbers into real integers and decimals.

Phase 3: Relational Modeling & Surrogate Key Normalization (Silver Star Schema)

Action: Normalizes your data by breaking giant text fields into independent lookup collections.

Format: Creates isolated entity arrays (`dim_players`, `dim_teams`, `dim_venues`) using fast, low-overhead auto-incrementing integer keys (`IDENTITY(1,1)`) linked directly to core transactional hub tables (`fact_matches`, `fact_deliveries`).

Phase 4: High-Performance Semantic Views Layer (Gold Data Marts)

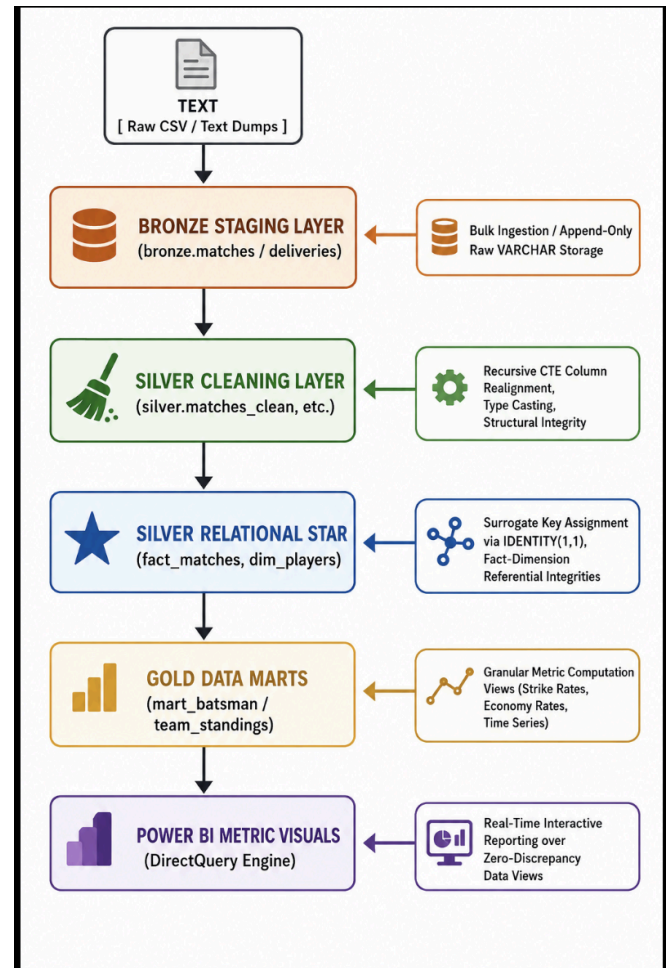
Action: Packages raw metrics inside clean SQL views designed specifically to serve analytical requests.

Format: Instantly computes high-level performance calculations including career Batting Strike Rates, Bowler Economy factors, and year-by-year historical Team Standings.

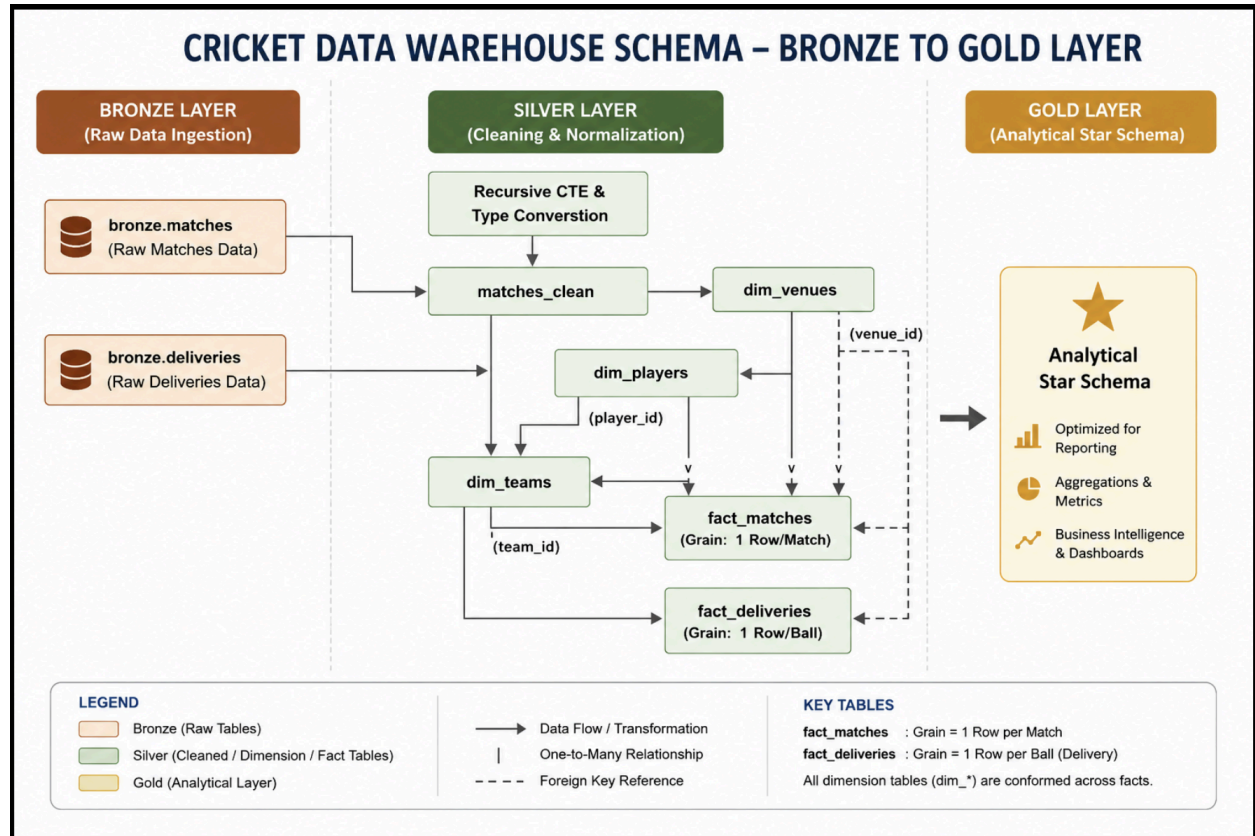
Phase 5: Interactive Enterprise Visualization (Power BI DirectQuery Service)

Action: Maps the Gold layer views into interactive corporate visuals like performance scatter matrices and leaderboard grids.

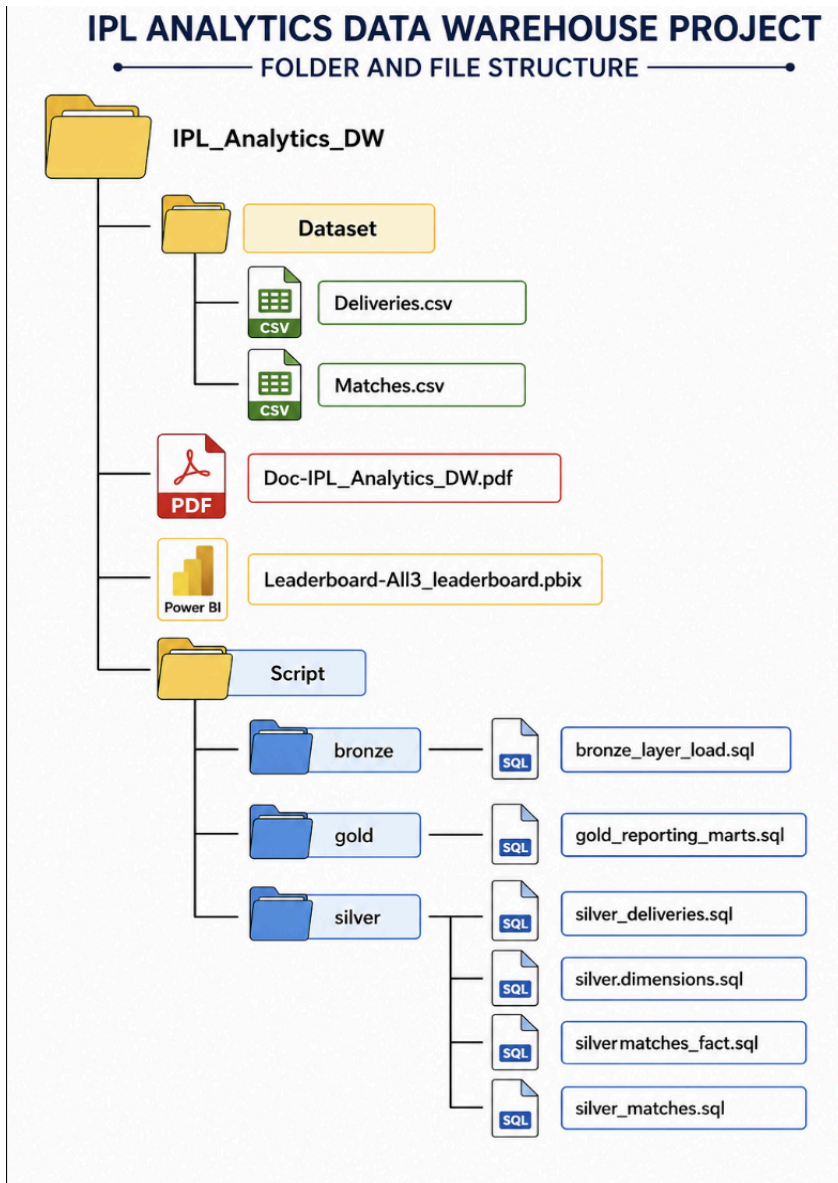
Format: Powered entirely by DirectQuery Mode. Because the database engine does all the heavy structural heavy lifting, Power BI acts as a fast visual wrapper with zero local memory overhe



2) Cricket Analytics ETL & Star Schema Architecture



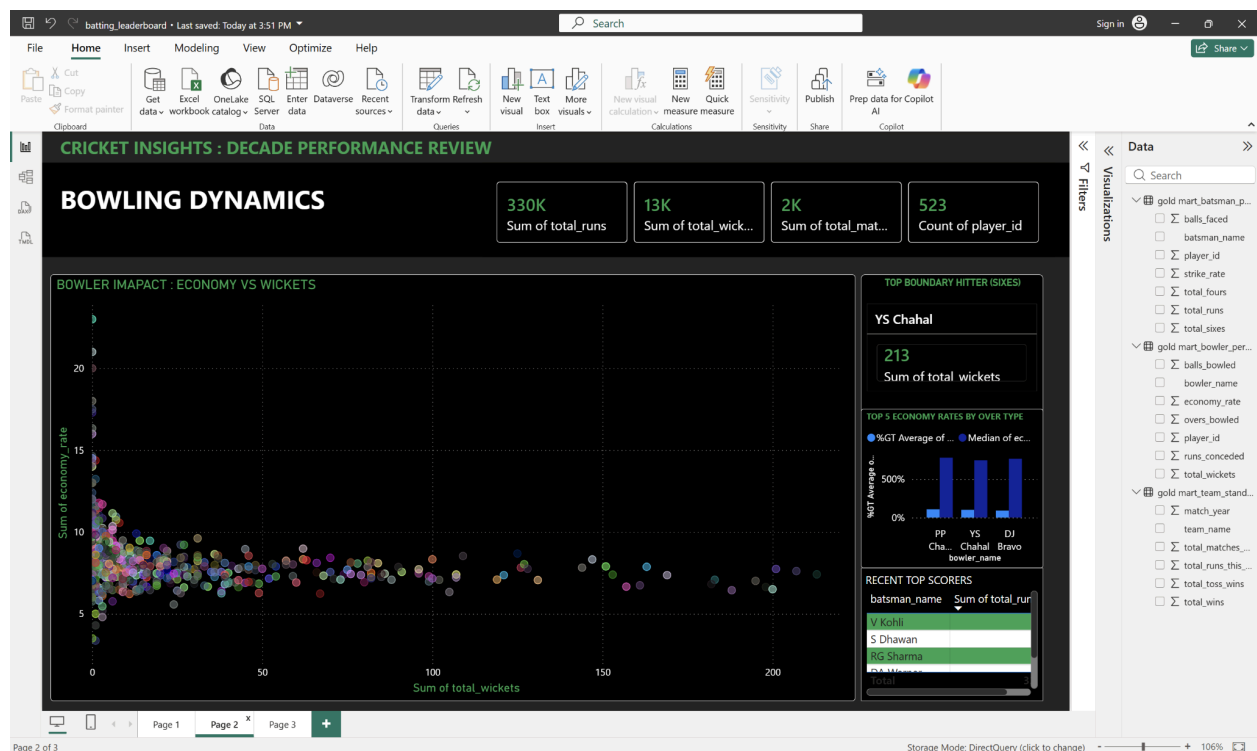
3) Folder and file Structure



3) Interactive Reporting & Visual Analytics Modules using POWER BI

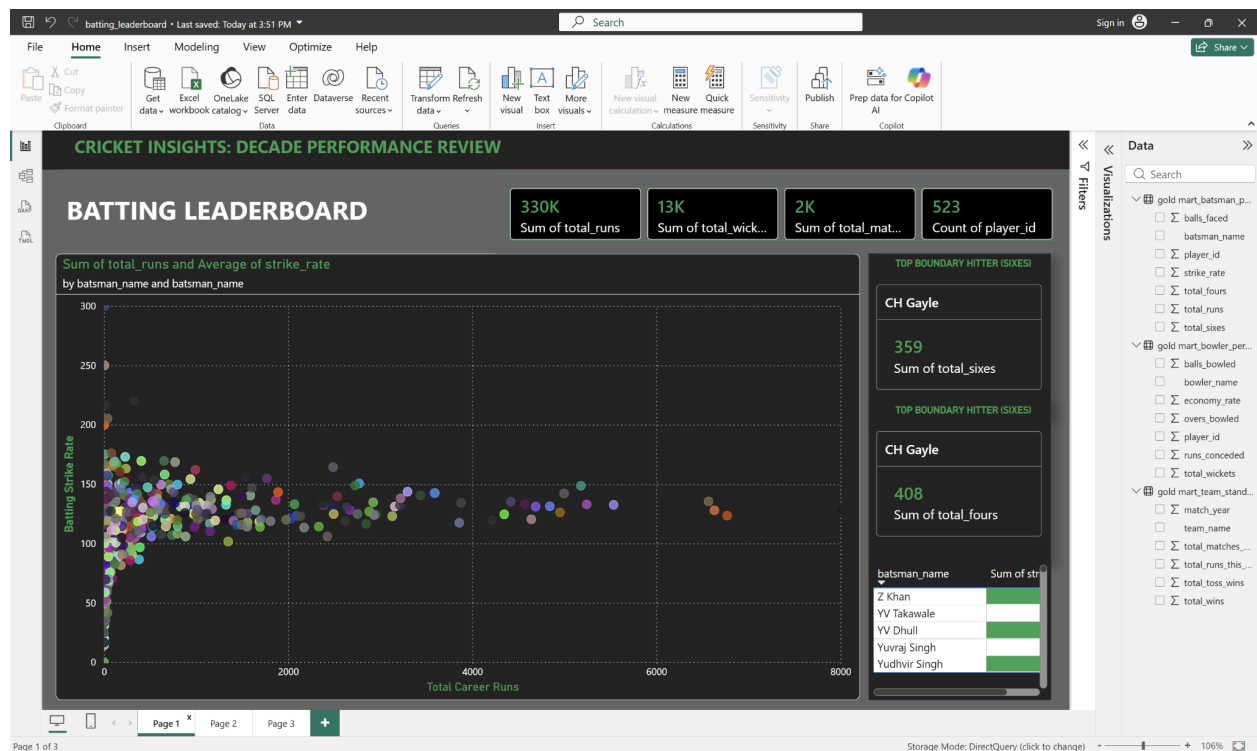
I. Defensive Bowling Efficiency Matrix

- **Title:** Bowling Restrictive Performance Scale (Economy vs. Wickets)
- **Subtitle:** Multi-Dimensional Efficiency Mapping for Defensive Spell Impact
- **Technical Documentation Points:**
 - Evaluates complex relational metrics (Economy Rates and Gross Wickets) offloaded directly onto the SQL Server engine.
 - Factors out extraneous extras and run penalties to analyze pure ball-by-ball spatial control.
 - Implements strict minimum overs constraints (≥ 6 balls bowled) to ensure highly reliable analytical data distributions.



II. Advanced Batting Trajectory Analysis

- **Title:** Striker Performance Matrix (Strike Rate vs. Gross Runs)
- **Subtitle:** High-Volume Outlier Identification and Scoring Profiles
- **Technical Documentation Points:**
 - Maps granular player calculations (Strike Rates vs. Total Runs) computed over millions of distinct database delivery rows.
 - Utilizes explicit visual coordinates to split individual player nodes cleanly without breaking coordinate groupings.
 - Features custom metric filters to isolate professional career thresholds and weed out low-volume part-time batters.



III .Dynamic Tournament Standings Matrix

- **Title:** Dynamic Tournament Standings Matrix
- **Subtitle:** DirectQuery Analytical View over Seasonal Performance Trends
- **Technical Documentation Points:**
 - Pulls live data streams from the database layer (`gold.mart_team_standings`) without local storage overhead.
 - Consolidates dynamic match outcomes, team wins, and toss metrics instantly across changing calendar years.
 - Employs real-time slicing filters (`match_year`) to allow on-the-fly franchise historical comparisons.

